

Seeed Studio XIAO ESP32C3

Power Consumption

- Equipment used for testing:
 - 1.1 Keysight N6705C, used to simulate battery power supply.



- 1.2 Testing DUT(Seeed Studio XIAO ESP32C3)



2. Test result

Connection status: WiFi: ☒ BT: ☐

	Normal Model	Modem Sleep	Light Sleep
Current	74mA	24mA	3mA

Connection status: WiFi: ☐ BT: ☒

	Normal Model	Modem Sleep	Light Sleep
Current	NO	24mA	9mA

Deep sleep mode: 43uA

3. WiFi Power Consumption Testing

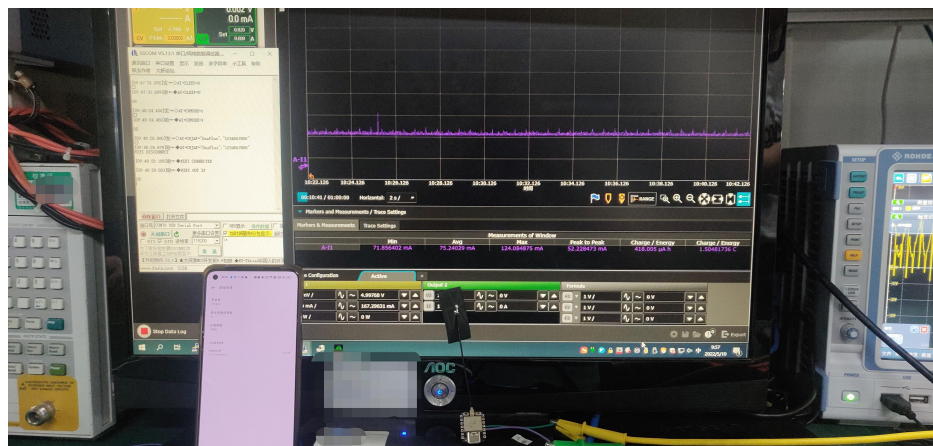
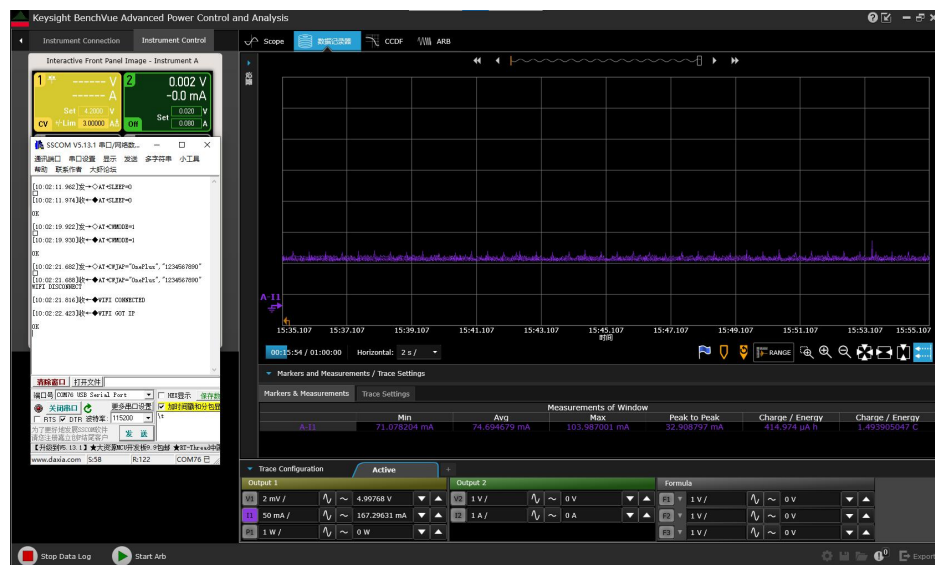
- a. Flash AT firmware.
- b. Sending the following command to DUT by serial port:

AT+CWMODE=1 -> Configure in Station mode.

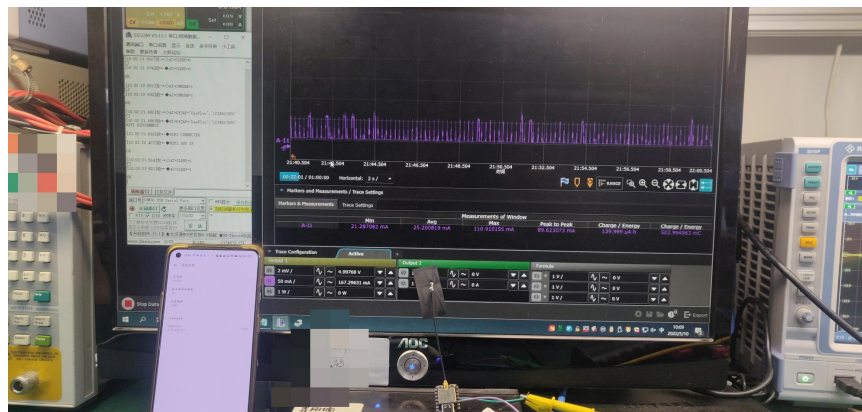
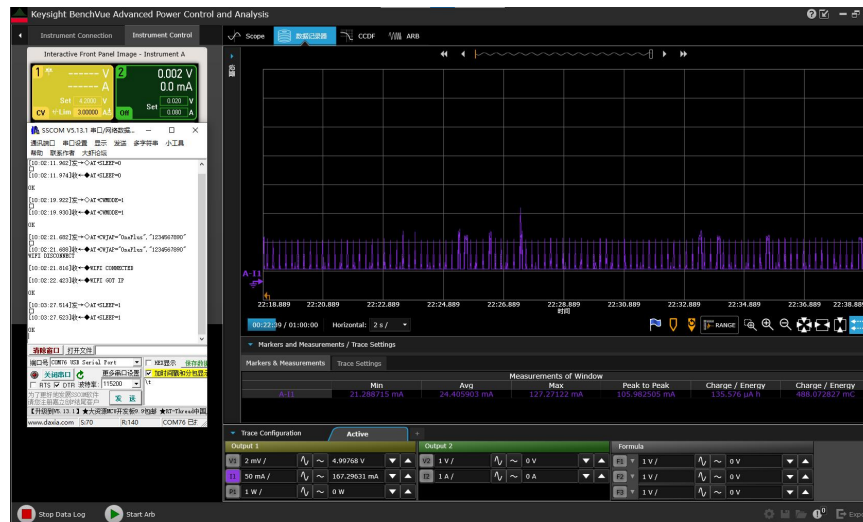
AT+CWJAP="SSID","Password" -> DUT will connect to the AP.

AT+SLEEP=0 -> Configure to normal power mode.

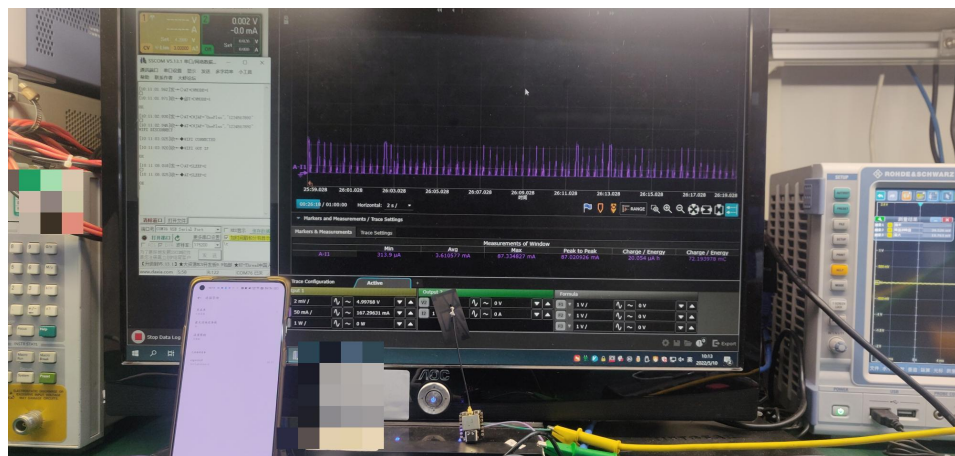
- c. Measure the current.



- d. Sending the following command to DUT through serial port
`"AT+SLEEP=1"` -> Configured DUT to modem sleep mode.
- e. Measure the current.



- f. Sending the following command to DUT through serial port
`AT+SLEEP=2` -> Configured DUT to light sleep mode.
- g. Measure the current.



4. Bluetooth power consumption Testing

- a. Sending the following command to DUT through serial port

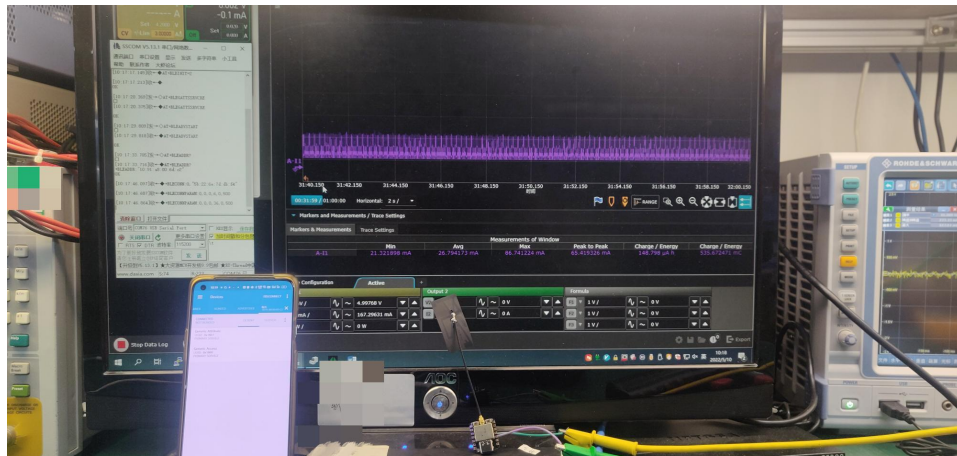
AT+BLEINIT=2 -> Configured DUT to Server mode.

AT+BLEADVSTART -> Turn on broadcast.

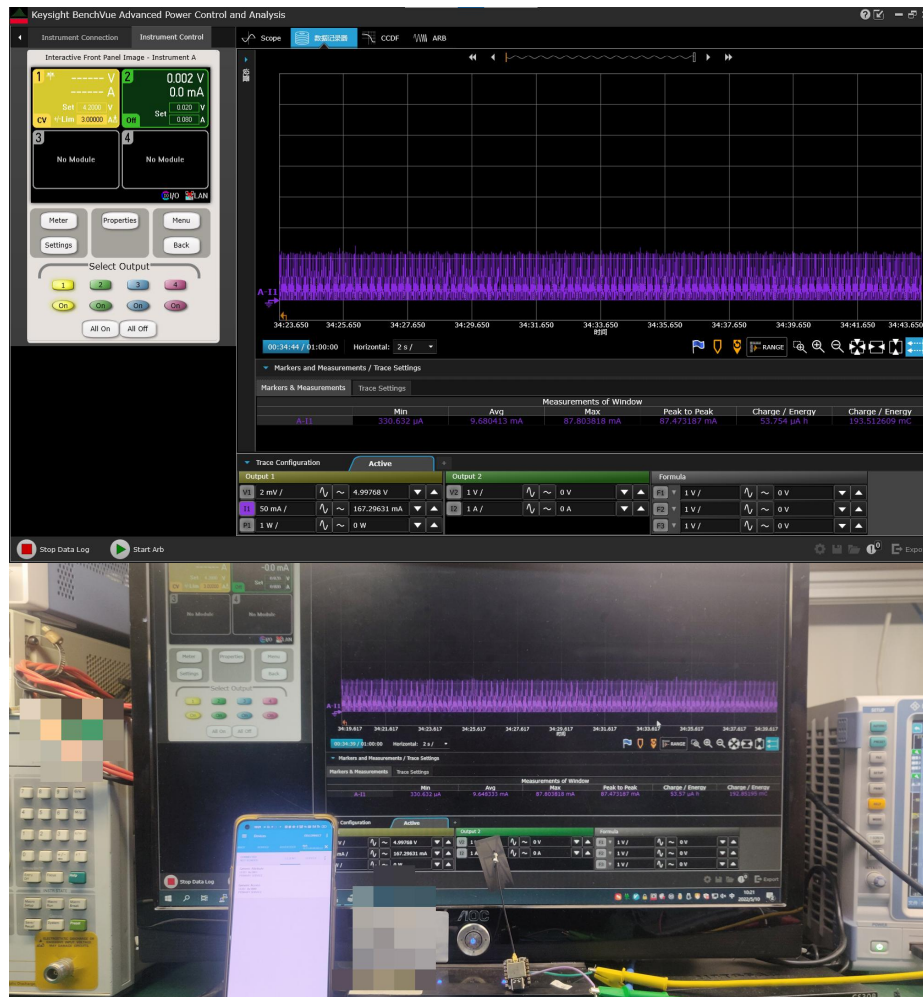
AT+CWMODE=0 -> Close WIFI RF.

AT+SLEEP=1 -> Configure DUT to modem sleep mode

- b. Measure the current.



- c. Sending the following command to DUT through serial port
AT+SLEEP=2 -> Configured DUT to light sleep mode.
- d. Measure the current.



5. Deep Sleep Mode Testing

a. Sending the following command to DUT through serial port

AT+GSLP=sleeptime

-> Configured DUT to deep sleep mode and keep sleeptime.

b. Measure the current.

